

1. Phân tích thuật toán (Algorithm Analysis)

- Khái niệm Big-O
- Master's Theorem

2. Sorting Algorithms

- Insertion Sort
- Merge Sort
- Quick Sort
- Heap Sort
- Counting Sort
- Radix Sort
- Bucket Sort

(Self-study)

- Selection Sort
- Bubble Sort

3. Design Techniques

Divide and Conquer

- Merge Sort (example)
- Median Finding

4. Graph Algorithms

a. Graph Traversal

- BFS (Breadth-First Search)
- DFS (Depth-First Search)

b. Disjoint Set Data Structure (Union-Find)

c. Minimum Spanning Tree

- Kruskal
- Prim

d. Shortest Path Algorithms

- Single Source Shortest Path
- All Pairs Shortest Path

5. Pattern Matching

- Knuth–Morris–Pratt (KMP)

6. Dynamic Programming

- Rod Cutting
- Matrix Chain Multiplication

7. Complexity Theory

- P
- NP
- NP-Complete
- NP-Hard
- Reduction

8. Approximation Algorithms

- Subset Sum
- Travelling Salesman Problem (TSP)